TOPIC 30/60

# What is Supervised Fine-Tuning?

> SFT: The crucial step that turns a wild text-completing calculator into a helpful, conversational AI.


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# THE WILD BASE MODEL

A raw "Base Model" (fresh out of pre-training) knows the internet, but it does not know how to act like an assistant. It only completes patterns.

### The Behavior:

**User:** "Write a poem about rain."

**Base LLM: "Write a poem about snow. Write a poem about wind."**

It completed your sentence template instead of answering your instruction.

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## ENTER SFT

# Behavior Alignment

Supervised Fine-Tuning (SFT) fixes this pattern loop. We feed the model thousands of high-quality, human-validated examples of high-tier conversations.

Instead of learning *grammar* or *everything about the world*, SFT teaches the model how to **format its outputs** and **obey user prompts**.

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THE TRAINING DATA

# The Dataset

SFT requires carefully structured high-quality instruction-response datasets.

```
{
  "prompt": "Explain gravity to a 10yo.",
  "response": "Gravity is like an invisible glue that keeps our feet on the ground..."
}
```

Thousands of clean, annotated pairs like this form the core curriculum of the SFT phase.

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UNDER THE HOOD

# Masked Loss

How does the model only learn the answers without repeating the prompts?

**Prompt:** [Masked - No Loss calculated]

**→ Response: [Calculate Loss & Updates]**

The math ignores prediction errors on the user's prompt, adjusting neural weights **only** to perfect the assistant's response.

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THE LIMA SHIFT

# Less is More

For a long time, companies thought SFT needed millions of conversational datasets. Then came the **LIMA hypothesis** (Meta research).

## 1,000 High-Quality Examples

Just 1,000 meticulously clean prompts matched with brilliant responses out-performed models aligned on 50,000 messy crowdsourced samples.

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# WHAT SFT TEACHES

STRUCTURE & OUTPUT FORMAT

### Reliable JSON/Markdown

SFT trains the model to structure code cleanly or output exact JSON fields required for developers.

TONE & BEHAVIOR

### Helpful & Polite Persona

It stops the model from insulting the user, adopting an eager, conversational, and polite assistant tone.

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THE LIMITATIONS

# Limits of SFT

While incredibly powerful, Supervised Fine-Tuning isn't a silver bullet for everything.

❌ **Sycophancy:** It learns to mimic human-validated text, often lying to agree with whatever the user says.

❌ **Hallucination:** It cannot reliably learn massive new raw facts through SFT—doing so leads to confidently wrong outputs.

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# PRE-TRAINING VS SFT

## Pre-training (Phase 1)

Billions of tokens. Learn grammar, code, history, and physical facts. High computational demand. Out: **Raw Base Model**.

## SFT / Alignment (Phase 2)

Thousands of clean chat records. Learn how to listen, format outputs, and serve the user. Low compute. Out: **Chat/Instruct Model**.

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# SFT

CONCEPT UNLOCKED

- Turns text predictors into conversational agents
- Emphasizes data quality over extreme volume
- Restricts learning updates only to response vectors
- Standardizes formatting like JSON & Markdown


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